

Performance Benchmarking of Embedded Battery State Estimation: A Measurement-Only Comparison of Direct-Measurement, Hybrid, and Data-Driven Estimators

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Abstract

This paper presents a measurement-only performance benchmark for embedded battery state estimation under realistic Battery Management System (BMS) constraints. The study isolates estimator performance rather than hardware-resource or implementation-effort evaluation and compares four embedded-ready estimators on the same Lithium Iron Phosphate (LFP) full-cell trajectory under a shared embedded constraint: a direct-measurement model (DM), a hybrid direct-measurement model with learned State-of-Health support (HDM), a hybrid equivalent-circuit model with voltage feedback (HECM), and a data-driven neural model with GRU-based SOC estimation and LSTM-based online SOH context (DD). The benchmark perturbs only measurable inputs and timing information, including ADC quantization, current bias, additive current, voltage, and temperature noise, missing samples, irregular sampling, burst dropout, voltage spikes, and initial-state mismatch, so that all disturbances remain physically interpretable and reproducible. The comparison reveals no universal winner across deployment-relevant dimensions. Under clean conditions, HDM provides the lowest baseline error with an MAE of 0.0024, followed by HECM with 0.0090,

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DD with 0.0100, and DM with 0.0311. Under disturbed measurements, HECM is the strongest overall robustness choice, especially in the most discriminative current-bias and voltage-noise scenarios. DD is the most stable estimator under missing samples and recovers fastest after initialization mismatch, while DM shows the expected drift sensitivity and HDM trades nominal accuracy against reduced disturbance tolerance. The benchmark therefore exposes complementary failure modes of direct-measurement, hybrid physics-based, and data-driven estimators and supports decision-oriented model selection for embedded BMS deployment.

Keywords: Battery management system, State of charge, State of health, Performance benchmark, Direct measurement, Equivalent circuit model, Data-driven estimation, Embedded artificial intelligence

1 Nomenclature

2 *Acronyms*

3	BMS	Battery management system
4	SOC	State of charge
5	SOH	State of health
6	DM	Direct-measurement model
7	HDM	Hybrid direct-measurement model
8	HECM	Hybrid equivalent-circuit model
9	DD	Data-driven neural model
10	ECM	Equivalent circuit model
11	EKF	Extended Kalman filter
12	MAE	Mean absolute error
13	RMSE	Root-mean-square error
14	ADC	Analog-to-digital converter
15	CV	Constant-voltage phase
16	OCV	Open-circuit voltage

17	GRU	Gated recurrent unit
18	LSTM	Long short-term memory
19	LFP	Lithium iron phosphate
20	<i>Symbols</i>	
21	t	Time
22	$I(t)$	Current signal at time t
23	$U(t)$	Voltage signal at time t
24	$T(t)$	Temperature signal at time t
25	$\widehat{\text{SOC}}(t)$	Estimated state of charge at time t
26	$\widehat{\text{SOH}}(t)$	Estimated state of health at time t
27	$\text{SOC}_{\text{ref}}(t)$	Reference SOC target at time t
28	$\text{SOH}_{\text{ref}}(t)$	Reference SOH target at time t
29	ΔI	Current bias or offset
30	ΔU	Voltage bias or offset
31	ΔT	Temperature bias or offset
32	σ_I	Standard deviation of injected current noise
33	σ_U	Standard deviation of injected voltage noise
34	σ_T	Standard deviation of injected temperature noise
35	Q_c	Online cumulative charge state used by the SOC models
36	C_{nom}	Nominal capacity
37	C_{eff}	Effective capacity used inside the estimator
38	U_{max}	Upper voltage threshold used for full-charge detection
39	U_{tol}	Voltage tolerance used in the full-charge condition
40	U_1, U_2	Polarization voltages of the first and second RC branch
41	ΔMAE	Error increase relative to baseline

42 1. Introduction

43 1.1. Motivation and practical relevance

44 Reliable battery state estimation is a core BMS function because SOC and
45 SOH determine safety margins, usable energy, operating limits, charge con-
46 trol, and degradation-aware system management in both mobile and stationary
47 lithium-ion battery applications [1, 2, 3]. Practical deployment quality is not de-
48 fined by clean-condition accuracy alone, because real BMS operation is exposed
49 to sensor bias, additive noise, initialization uncertainty after restart, missing
50 samples, irregular sampling, communication interruptions, and transient signal
51 spikes caused by measurement-chain limitations or disturbances in the surround-
52 ing system [4, 5, 6]. The relevant engineering question is therefore not only
53 whether an estimator achieves low MAE under nominal measurements, but also
54 whether it converges back to a reliable state after disturbed initialization and
55 whether it remains stable under physically plausible measurement corruption.

56 As summarized conceptually in Figure 1, the present benchmark treats
57 deployment-facing performance as a three-dimensional problem consisting of
58 nominal accuracy, convergence behavior after state inconsistency, and robust-
59 ness against disturbed measurements and signal-integrity faults, embedded within
60 broader BMS requirements such as hardware constraints and implementation
61 effort. The present study therefore isolates the performance branch and com-
62 pares representative estimator classes under identical measurement-side distur-
63 bances and a matched embedded-feasibility constraint. The benchmark shows
64 that strong nominal accuracy does not necessarily imply strong disturbance
65 performance and that the estimator classes respond differently to bias, missing
66 samples, and re-initialization events.

67 1.2. State of the art in battery state estimation

68 The literature commonly groups battery-state estimators into direct-measurement
69 or integration-based approaches, model-based approaches based on equivalent-
70 circuit or electrochemical models, hybrid approaches that combine physical

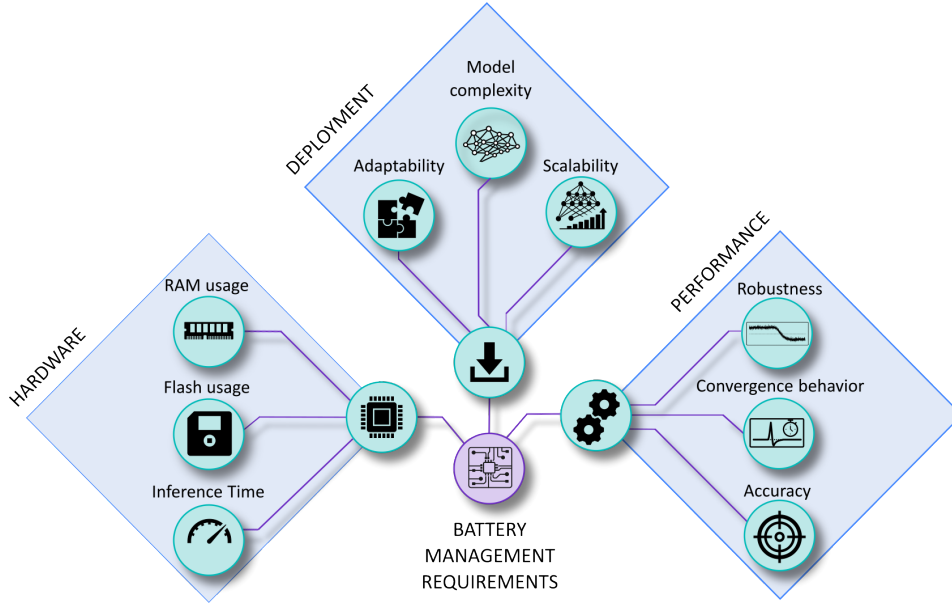


Figure 1: Conceptual overview of deployment-oriented requirements for embedded BMS state estimation. The broader requirement space includes implementation effort, hardware constraints, and estimator performance, whereas the present study focuses on the performance branch and evaluates baseline accuracy, convergence behavior, and robustness under measurement disturbances.

71 structure with learned components, and purely data-driven models based on
 72 neural sequence learning [7, 1, 2, 3]. Direct-measurement approaches, typically
 73 based on Coulomb counting, are attractive because they are simple, computa-
 74 tionally inexpensive, and easy to deploy; however, they remain structurally sen-
 75 sitive to current bias, missing data, and initialization errors because they lack an
 76 internal correction mechanism [7, 4]. Model-based approaches, especially ECM-
 77 and observer-based estimators, offer physical interpretability and voltage-based
 78 correction capability, but their practical behavior depends strongly on the qual-
 79 ity of model parametrization, operating-point observability, and measurement
 80 quality [8, 2, 4].

81 Hybrid estimators seek to combine physical state propagation with learned
 82 adaptation of capacity, parameters, or auxiliary states, which has made joint

83 SOC/SOH estimation an active research direction [9, 10, 11]. This combination
 84 can improve baseline calibration, while it does not necessarily eliminate struc-
 85 tural sensitivity to measurement disturbances. Data-driven approaches based
 86 on recurrent or sequence models can exploit nonlinear cross-relations between
 87 current, voltage, temperature, and derived online features, and recent work has
 88 also demonstrated their embedded feasibility through TinyML-oriented imple-
 89 mentations, pruning, and quantization [12, 13, 14, 15, 3].

90 *1.3. Performance and embedded deployment gap*

91 A large share of the SOC estimation literature still emphasizes clean-data ac-
 92 curacy, average error metrics, or computational efficiency, while comparatively
 93 fewer studies analyze how estimators behave under realistic measurement faults
 94 and timing corruption [1, 13, 14, 15]. Existing robustness studies often remain
 95 confined to a single model family, for example observer-based robustness against
 96 modeling and measurement errors [4] or robust estimation concepts for specific
 97 BMS architectures under noise and uncertainty [5], whereas cross-family com-
 98 parisons under the same disturbance protocol and under comparable embedded
 99 constraints remain limited. Benchmarking studies in the broader BMS con-
 100 text underline the importance of reproducible scenario design and deployment-
 101 relevant validation, but they do not by themselves close the gap between clean-
 102 data estimator comparison and measurement-level performance benchmarking
 103 across fundamentally different estimator classes [6]. This gap is especially rele-
 104 vant in embedded BMS design, where estimator selection must jointly consider
 105 nominal accuracy, convergence, robustness under disturbed measurements, and
 106 computational constraints.

107 *1.4. Contribution and study objectives*

108 This work defines deployment-facing estimator performance as the joint as-
 109 sessment of nominal accuracy, convergence behavior after state inconsistency,
 110 and robustness under realistic disturbances applied only to measurable inputs
 111 and their timing. The study compares four embedded-ready representatives

of major estimator families, namely a direct-measurement model (DM), a hybrid direct-measurement model with learned SOH support (HDM), a hybrid equivalent-circuit model with voltage feedback (HECM), and a data-driven neural model with stateful GRU-based SOC estimation and stateful LSTM-based online SOH context (DD), under identical online execution conditions. The disturbances are deliberately restricted to measurement-level and signal-integrity scenarios, including current bias, current noise, voltage noise, temperature noise, missing samples, irregular sampling, burst dropouts, voltage spikes, ADC quantization, and initialization errors, so that the benchmark remains physically interpretable and experimentally reproducible.

The manuscript should therefore be read as a controlled cross-class estimator benchmark and not as a complete system-level BMS validation campaign. Its role is to isolate estimator-side failure modes under matched data, matched disturbance definitions, matched online execution rules, and matched embedded-feasibility constraints. The resulting contribution is a measurement-only benchmark that separates baseline accuracy, convergence, and disturbed-operation robustness and translates the observed failure modes into decision-oriented guidance for embedded BMS estimator selection.

2. Methodology

2.1. Estimator classes under test

Figure 2 summarizes the causal benchmark chain, from measured input signals and online feature reconstruction to disturbance injection and the final global and local performance evaluation. The direct-measurement model (DM) represents the canonical embedded Coulomb-counting approach with fixed nominal capacity and explicit full-charge reset logic, and serves as the low-complexity reference for integration-driven SOC estimation. All SOH-dependent estimator variants in the benchmark, namely HDM, HECM, and DD, use the same shared LSTM-based online SOH estimator; the classes differ only in how this common SOH signal is consumed inside the SOC pipeline. The hybrid direct-

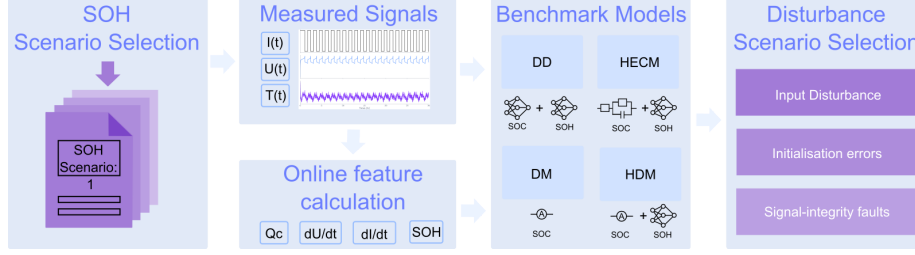


Figure 2: Methodology overview for the causal benchmark chain. Measured battery signals are converted into online auxiliary features, processed by representative estimator classes, exposed to measurement-only disturbances, and finally evaluated through global and local performance metrics.

141 measurement model (HDM) preserves the same Coulomb-counting core but re-
 142 places the fixed capacity assumption with an online effective capacity derived
 143 from this shared SOH estimator, which makes it representative of lightweight
 144 hybrid estimation. The hybrid equivalent-circuit model (HECM) represents
 145 physics-based state estimation with observer correction, because it combines
 146 current-driven state propagation with voltage-residual feedback and SOH-dependent
 147 parameterization in a second-order RC equivalent-circuit structure driven by
 148 the same shared SOH estimate [8, 2]. The data-driven model (DD) represents
 149 a neural estimator pipeline in which SOC is predicted by a lightweight stateful
 150 GRU-based sequence model while the shared LSTM-based online SOH estimate
 151 is provided as causal context under the same execution constraints [12, 3].

152 2.2. SOC estimation methods

153 2.2.1. Direct-measurement model (DM)

154 The direct-measurement SOC formulation used in DM, HDM, and as an
 155 explicit auxiliary feature in DD follows the same online cumulative-charge state,
 156 denoted here uniformly as Q_c . It is updated causally from the measured current
 157 stream as

$$Q_c(k) = Q_c(k-1) + \frac{I_k \Delta t_k}{3600 \text{ s h}^{-1}}, \quad (1)$$

158 where I_k is given in ampere and Δt_k in seconds, so that the factor 3600 s h^{-1}
 159 converts the increment to ampere-hours. The sign convention is fixed by the

benchmark implementation and the state is reset to $Q_c = 0$ after a sustained constant-voltage full-charge event. In the DM, this same charge state is mapped directly to SOC through the fixed nominal capacity

$$\widehat{\text{SOC}}_{\text{DM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{nom}}}, 0, 1\right), \quad (2)$$

2.2.2. Hybrid direct-measurement model (HDM)

The HDM preserves exactly the same Coulomb-counting core and differs only in the capacity term. The SOH estimate rescales the usable capacity as

$$C_{\text{eff}}(k) = C_{\text{nom}} \cdot \widehat{\text{SOH}}(k), \quad (3)$$

which yields

$$\widehat{\text{SOC}}_{\text{HDM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{eff}}(k)}, 0, 1\right). \quad (4)$$

This formulation makes the relation between DM and HDM explicit: both use the same online charge state Q_c , while only the denominator changes through online SOH support. In both direct-measurement variants, a sustained near-maximum voltage condition is interpreted as a constant-voltage full-charge event; if the voltage remains above $U_{\text{max}} - U_{\text{tol}}$, where U_{tol} denotes the admissible voltage margin below the upper threshold U_{max} , for the configured CV duration, the internal charge state is reset so that $Q_c = 0$ and therefore $\widehat{\text{SOC}} = 1$, which reflects the practical assumption that a completed CV phase provides a physically anchored full-charge reference.

2.2.3. Hybrid equivalent-circuit model (HECM)

The hybrid ECM model is implemented as a second-order RC model with the state vector

$$\mathbf{x}_k = \begin{bmatrix} \widehat{\text{SOC}}_k & U_{1,k} & U_{2,k} \end{bmatrix}^T, \quad (5)$$

where U_1 and

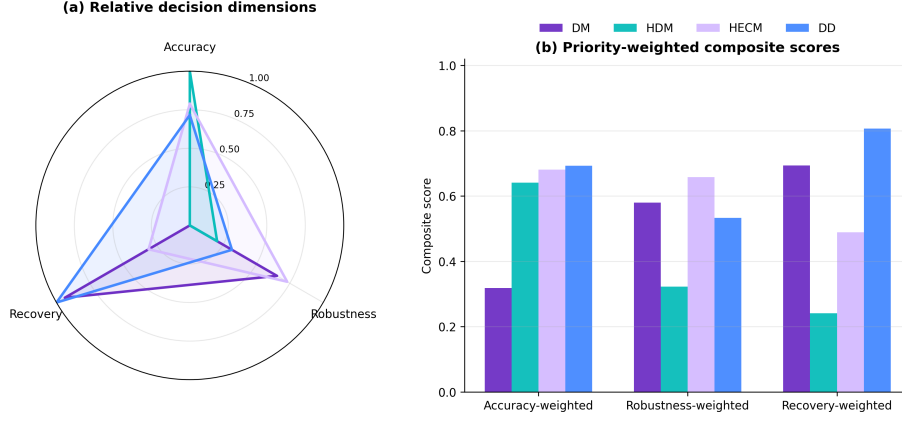


Figure 13: Decision-oriented synthesis of the benchmark results. Panel (a) collapses the benchmark into three relative, min-max-normalized decision dimensions, namely nominal accuracy, disturbed-scenario robustness, and recovery behavior. Panel (b) shows three illustrative deployment-priority profiles and is intended only as a visual complement to Table 4, not as a replacement for the raw scenario metrics.

hardware-in-the-loop or true embedded runtime benchmarks under the same performance-oriented evaluation philosophy.

6. Conclusion

This work frames battery-state estimator evaluation as a performance problem under realistic measurement corruption rather than as a clean-data accuracy comparison alone. The central engineering message is that estimator selection depends on the expected disturbance profile of the BMS measurement chain, the required recovery behavior after corrupted inputs, and the practical implementation constraints of the target system, because no single estimator dominates all deployment-relevant dimensions. The main findings can be summarized by estimator family: the hybrid direct-measurement model is the nominal-accuracy leader, the hybrid equivalent-circuit model with shared data-driven SOH support is the strongest overall robustness choice in the most discriminative disturbance scenarios, and the data-driven neural model is strongest in recovery after initialization mismatch and in missing-sample resilience.

620 If one general-purpose estimator must be preferred under the specific distur-
 621 bance profile emphasized here, the benchmark supports the hybrid equivalent-
 622 circuit approach as the strongest overall compromise. Its combination of in-
 623 terpretable physical parameterization, voltage-based correction, and compact
 624 shared SOH support preserves a clear role for ECM-based estimation in embed-
 625 ded BMS design. At the same time, the data-driven model class remains highly
 626 attractive when the development strategy includes iterative data collection and
 627 model updates. The present benchmark also yields a practical design lesson:
 628 data-driven SOC estimators should not rely too strongly on one dominant trig-
 629 ger variable, but should be trained and feature-engineered to make balanced
 630 use of the available current, voltage, temperature, and health context if they
 631 are to realize their full robustness potential. The benchmark is designed to be
 632 extensible, and the next steps are the extension to more cells and scenarios, the
 633 inclusion of remaining disturbance classes, and the transfer of the methodology
 634 toward broader embedded or hardware-in-the-loop validation.

635 Appendix A. Supplementary Result Tables

Table A.5: Baseline performance metrics for the selected bench-
mark trajectory.

Class	MAE	RMSE	P95 error	Max error	Bias
Direct measurement	0.0311	0.0397	0.0800	1.0000	0.0311
Hybrid direct measurement	0.0024	0.0061	0.0094	1.0000	0.0022
Hybrid ECM	0.0090	0.0127	0.0241	1.0000	-0.0029
Data-driven	0.0100	0.0143	0.0336	0.0686	-0.0062

Table A.6: Cross-scenario global performance metrics used in the result discussion.

Scenario	Class	MAE	Δ MAE	RMSE	Δ RMSE	P95 error
Current noise (high)	DM	0.0310	-0.0001	0.0394	-0.0002	0.0795
Current bias	DM	0.3143	0.2833	0.3683	0.3286	0.6670
Initial SOC error	DM	0.0322	0.0011	0.0413	0.0017	0.0833
Irregular sampling	DM	0.0309	-0.0001	0.0396	-0.0001	0.0799
Burst dropout	DM	0.0332	0.0022	0.0487	0.0090	0.0819
Missing samples	DM	0.0386	0.0076	0.0476	0.0079	0.0926
Voltage spikes	DM	0.0312	0.0001	0.0398	0.0001	0.0801
Temperature noise	DM	0.0311	0.0000	0.0397	0.0000	0.0800
Voltage noise	DM	0.0524	0.0213	0.0648	0.0252	0.1228
Current noise (high)	HDM	0.0044	0.0019	0.0074	0.0013	0.0120
Current bias	HDM	0.3068	0.3043	0.3638	0.3577	0.6668
Initial SOC error	HDM	0.0036	0.0011	0.0136	0.0075	0.0101
Irregular sampling	HDM	0.0027	0.0003	0.0061	0.0001	0.0095
Burst dropout	HDM	0.0051	0.0027	0.0325	0.0265	0.0098
Missing samples	HDM	0.0103	0.0079	0.0129	0.0069	0.0221
Voltage spikes	HDM	0.0062	0.0038	0.0140	0.0079	0.0328
Temperature noise	HDM	0.0067	0.0043	0.0110	0.0049	0.0210
Voltage noise	HDM	0.0751	0.0727	0.0908	0.0847	0.1578
Current noise (high)	HECM	0.0165	0.0075	0.0280	0.0154	0.0701
Current bias	HECM	0.0932	0.0842	0.1303	0.1176	0.2954
Initial SOC error	HECM	0.0091	0.0001	0.0143	0.0016	0.0241
Irregular sampling	HECM	0.0095	0.0004	0.0131	0.0004	0.0248
Burst dropout	HECM	0.0104	0.0014	0.0228	0.0101	0.0246
Missing samples	HECM	0.0119	0.0029	0.0163	0.0036	0.0329
Voltage spikes	HECM	0.0090	0.0000	0.0127	-0.0000	0.0241
Temperature noise	HECM	0.0090	0.0000	0.0127	0.0000	0.0241

Scenario	Class	MAE	Δ MAE	RMSE	Δ RMSE	P95 error
Voltage noise	HECM	0.0090	-0.0000	0.0127	-0.0000	0.0241
Current noise (high)	DD	0.0109	0.0009	0.0151	0.0008	0.0345
Current bias	DD	0.2859	0.2759	0.3426	0.3283	0.6367
Initial SOC error	DD	0.0110	0.0010	0.0175	0.0032	0.0355
Irregular sampling	DD	0.0105	0.0005	0.0144	0.0001	0.0330
Burst dropout	DD	0.0122	0.0022	0.0316	0.0173	0.0348
Missing samples	DD	0.0117	0.0016	0.0149	0.0006	0.0302
Voltage spikes	DD	0.0144	0.0044	0.0213	0.0070	0.0474
Temperature noise	DD	0.0181	0.0081	0.0223	0.0080	0.0451
Voltage noise	DD	0.0698	0.0597	0.0839	0.0696	0.1501

Table A.7: Local behavior and recovery metrics used in the result discussion.

Class	Scenario	Local metric	Value	Threshold
DM	Initial SOC error	Recovery time to strict band [h]	1.1658	0.0800
DM	Initial SOC error	Recovery time to fair band [h]	0.2500	0.1599
HDM	Initial SOC error	Recovery time to strict band [h]	–	0.0094
HDM	Initial SOC error	Recovery time to fair band [h]	–	0.0200
HECM	Initial SOC error	Recovery time to strict band [h]	2.3693	0.0241
HECM	Initial SOC error	Recovery time to fair band [h]	1.1564	0.0482
DD	Initial SOC error	Recovery time to strict band [h]	1.0464	0.0336
DD	Initial SOC error	Recovery time to fair band [h]	1.0014	0.0672
DM	Burst dropout	Recovery time after burst dropout [h]	27.3511	0.0216
HDM	Burst dropout	Recovery time after burst dropout [h]	27.4093	0.0008
HECM	Burst dropout	Recovery time after burst dropout [h]	28.5611	0.0095
DD	Burst dropout	Recovery time after burst dropout [h]	27.4351	0.0070
DM	Voltage spikes	Median spike recovery time [s]	0.0000	0.0800

Class	Scenario	Local metric	Value	Threshold
HDM	Voltage spikes	Median spike recovery time [s]	0.0000	0.0094
HECM	Voltage spikes	Median spike recovery time [s]	0.0000	0.0241
DD	Voltage spikes	Median spike recovery time [s]	0.0000	0.0336
DM	Current noise (high)	Late minus early excess rolling MAE	-0.0013	–
DM	Current noise (high)	Mean excess rolling MAE	-0.0001	–
HDM	Current noise (high)	Late minus early excess rolling MAE	-0.0011	–
HDM	Current noise (high)	Mean excess rolling MAE	0.0019	–
HECM	Current noise (high)	Late minus early excess rolling MAE	0.0038	–
HECM	Current noise (high)	Mean excess rolling MAE	0.0073	–
DD	Current noise (high)	Late minus early excess rolling MAE	-0.0002	–
DD	Current noise (high)	Mean excess rolling MAE	0.0009	–

Table A.8: Scenario overview linking disturbance configuration, primary evidence type, and main paper figure. This table clarifies which scenarios are interpreted predominantly through global metrics and which require additional local evidence.

Scenario	Configuration	Primary evidence	Main figure(s)
ADC quantization	Steps of 0.01 A, 0.005 V, and 0.5 °C	Global Δ MAE; detailed values in Appendix	Figs. 12, B.14
Current bias	Constant current offset in the compact matrix, plus dedicated 0.5%/1.5%/3.0% sweep of $ I _{\max}$	Global MAE degradation across the sweep	Fig. 5

Scenario	Configuration	Primary evidence	Main figure(s)
Current noise	Compact matrix with $\sigma_I = 0.10$ A; local detail sweep up to $\sigma_I = 0.20$ A	Global Δ MAE plus local output-variability / rolling-MAE evidence	Fig. 6
Voltage noise	Additive Gaussian noise with $\sigma_U = 0.01$ V	Global Δ MAE	Fig. 12
Temperature noise	Additive Gaussian noise with $\sigma_T = 1.0$ °C	Global Δ MAE	Fig. 12
Initial SOC error	Additive initial mismatch of -0.10 SOC; DD via online Q_c offset	Local recovery to strict and fair baseline bands; global MAE is secondary	Fig. 7
Missing samples	Every 50th sample frozen on the nominal 1 s grid	Global Δ MAE	Figs. 8, 12
Irregular sampling	Uniform sampling-time jitter of ± 0.1 s around the nominal 1 s grid	Global Δ MAE	Figs. 8, 12
Burst dropout	One central frozen gap of 3600 s	Global Δ MAE plus local transition and long-horizon recovery	Figs. 8, 9, 10
Voltage spikes	One ± 0.20 V single-sample spike every 1000 samples	Global Δ MAE plus aligned local excess-error response	Figs. 11, 12

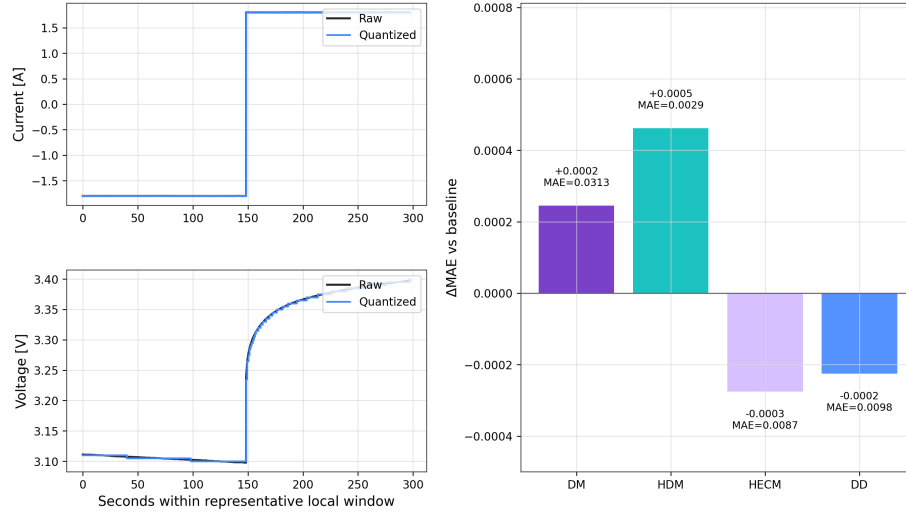


Figure B.14: ADC-quantization results. The two left panels illustrate representative current and voltage quantization in a local window, and the right panel summarizes the resulting global ΔMAE relative to the baseline.

Appendix B. ADC Quantization

ADC quantization is evaluated with step sizes of 0.01 A for current, 0.005 V for voltage, and 0.5 °C for temperature. Across all four estimator classes, the resulting deviations from the baseline remain small, so the detailed numerical comparison is collected here in the appendix while the scenario itself remains part of the overall benchmark.

Table B.9: ADC-quantization results for the four benchmarked estimator classes.

Class	Baseline MAE	ADC MAE	ΔMAE
DM	0.0311	0.0313	+0.0002
HDM	0.0024	0.0029	+0.0005
HECM	0.0090	0.0087	-0.0003
DD	0.0100	0.0098	-0.0002

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653 **Florian Rzepka:** Conceptualization, methodology, software, validation,
654 formal analysis, investigation, data curation, writing—original draft preparation,
655 writing—review and editing, visualization.

656 **Qinnan Chen:** Writing—review and editing.

657 **Julia Kowal:** Resources, supervision, project administration, funding acquisition,
658 writing—review and editing.

659 **Declaration of competing interest**

660 The authors declare that they have no known competing financial interests or
661 personal relationships that could have appeared to influence the work reported
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663 **Declaration of generative AI and AI-assisted technologies in the manuscript 664 preparation process**

665 During the preparation of this work the author(s) used ChatGPT, DeepL,
666 and Grammarly in order to improve the linguistic quality and style of the

manuscript. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Data availability

The underlying laboratory dataset is the MGFarm LFP 18650 cyclic-aging dataset described in [28]. The benchmark scripts and model runners are located in `DL\_Models/LFP\_SOC\_SOH\_Model/4\_simulation\_environment`, and the paper-facing figures and tables are collected under `DL\_Models/LFP\_SOC\_SOH\_Model/4\_simulation\_environment/results`. The data and code underlying this study will be made available on reasonable request.

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